

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India — Expected Aug. 2026

CGPA: 8.19 / 10.0

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS, MATLAB, Bash

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, Segment Anything (SAM), GANs

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, GraphQL, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Git/GitHub, Linux, Firebase, RAG Pipelines, Vercel

Research Experience

Undergraduate Researcher

June 2023 – Aug. 2023

Supervised by Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline to optimize **Segment Anything Model (SAM)** for agricultural crop-weed segmentation, improving mean IoU from 0.67 to 0.75 (12% improvement) on sparse datasets ($N = 500$ images).
- Engineered custom image enhancement techniques including contrast adjustment, noise reduction, and edge detection to maximize SAM's zero-shot segmentation performance on novel agricultural imagery.
- Reduced computer vision pipeline runtime from 8.5 hours to 5.1 hours (40% speedup) by implementing vectorized NumPy operations and multiprocessing for batch processing 10,000+ image datasets.

Industry Experience

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Collaborated with one other intern to build **FastAPI** backend processing 50,000+ lines of code daily across 15+ programming languages using Abstract Syntax Tree (AST) analysis, reducing engineering team's documentation time by an estimated 60%.
- Implemented hash-based change detection system for **vector database updates**, chunking codebases function-by-function and tracking content hashes to skip unchanged functions – reducing embedding generation costs by 85% by avoiding redundant processing.
- Developed 12 **RESTful API endpoints** with p95 latency under 200ms, designing response schemas to enable real-time code documentation preview in web interface.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs | Noida

- Refactored 15,000-line monolithic **RoboGPT** codebase into modular architecture with 40% fewer inter-module dependencies, enabling parallel development across 3 teams and reducing average bug fix time from 2 days to 6 hours.
- Integrated Computer Vision capabilities into robotic control loop, optimizing calibration routines and improving robotic arm positioning accuracy for complex food preparation tasks.

Selected Technical Projects

VR LLM Conversational Agent | Unity, C#, Gemini 2.5 Flash, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud Speech-to-Text and Text-to-Speech, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.
- Designed **within-subjects experimental study** ($N = 18$) to evaluate three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) on user perception using Godspeed questionnaire and custom metrics.
- Demonstrated through **statistical analysis (Wilcoxon signed-rank test, $p < 0.05$)** that embodied gestures significantly improved perceived animacy and immersion compared to explicit loading indicators.

- **Co-authored research paper** (in preparation for submission to Presence/India HCI/TOCHI) analyzing component-level latency feedback mechanisms in embodied AI agents, presented findings at university research symposium to 50+ attendees.

Reddit Data Ingestion Pipeline | *Python, PRAW, VADER, LDA, Firebase*

Aug. 2025 – Dec. 2025

- Built automated data pipeline using **PRAW** to scrape 22,000+ interactions from 19 pregnancy-related subreddits, implementing concurrent scraping with rate limiting and error recovery for 99.5% data completeness.
- Designed **Firebase NoSQL schema** with real-time deduplication logic using comment IDs and timestamps as composite keys, preventing duplicates across 5 parallel scraper instances.
- Discovered 19.4% sentiment shift ("Buffer Effect") in pre-eclampsia discussions using **VADER sentiment analysis** and **LDA topic modeling**, identified 35.8x volume increase post-pandemic indicating growing health anxiety in online communities.

Resume Builder & Interview GPT | *Next.js, Tailwind, Firebase, Gemini API*

Jan. 2025 – April 2025

- Built full-stack AI resume builder serving 500+ users with real-time PDF preview, generating ATS-optimized resumes and achieving 4.2/5 average rating from 80+ user submissions.
- Integrated **Gemini 2.5 Flash API** to power "Interview GPT" feature generating 1,000+ personalized DSA questions with detailed solutions, reducing average user preparation time by 40% based on feedback surveys.
- Implemented **Firebase Authentication** with Google OAuth supporting 200+ active users, designed responsive UI with Tailwind CSS for mobile and desktop experiences.

UI Wireframe Generator | *Python, PyTorch, Meta LLaMA, Computer Vision*

Aug. 2024 – Dec. 2024

- Explored generative approaches for UI wireframe synthesis from text prompts using **RICO dataset** (72,000+ UI screens), experimenting with Cross-Attention GAN (CAGAN) modified from SAGAN architecture.
- Fine-tuned **LLaMA-7B model** on custom text-UI pairs to translate natural language requirements into structured JSON layouts, ultimately achieving best results with LLM-based approach over diffusion models due to GPU constraints.
- Benchmarked multiple generative architectures (DDPM, Stable Diffusion, StyleGAN, SAGAN, CAGAN, LLaMA) documenting tradeoffs in quality, training stability, and compute requirements for UI generation tasks.

Kelp Forest Segmentation | *Python, PyTorch, U-Net, Satellite Imagery*

Jan. 2024 – May 2024

- Achieved **Top 100 ranking (83rd percentile)** in Kaggle satellite imagery competition using **U-Net with EfficientNet-B3** backbone for semantic segmentation of kelp forests in noisy satellite imagery.
- Engineered preprocessing pipeline using custom **CNN filters** to detect and remove streaky atmospheric artifacts from satellite images, improving model training stability and final segmentation quality.

C-VRP with Time Window Constraints | *Python, Google OR-Tools, Google Maps API*

Sept. 2023 – Dec. 2023

- Implemented optimization solution for **Capacitated Vehicle Routing Problem (VRP)** with time window constraints using **Google OR-Tools**, generating efficient routing solutions for multi-stop delivery scenarios.
- Integrated real-world transit data via **Google Distance Matrix API** to generate practical routing solutions, developed interactive visualizations using Google Maps API for route analysis and planning.

Leadership & Involvement

Head of Tech

April 2025 – Present

Athleda (Plaksha Sports Society) | Plaksha University

- Lead 4-person dev team building sports platform serving 500+ students with event registration, live scoring, and photo galleries – established CI/CD pipeline reducing deployment time from 45 minutes to 8 minutes.
- Conduct weekly code reviews and architectural planning sessions, mentored 2 junior developers in React best practices and Git workflows resulting in 60% reduction in production bugs.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Plaksha University

- Managed 6-person web team building AI club website attracting 800+ monthly users for events, workshops, and project showcases – coordinated 3 major redesigns based on user feedback.
- Mentored 6 junior developers in modern web stack (Next.js, TypeScript, Vercel), established team Git workflow reducing merge conflicts by 70% and onboarded new members with custom training curriculum.